

AETHEA

Hips & Pelvis Fracture Detection

Model Comparison: Faster R-CNN vs YOLOv8

FASTER R-CNN | YOLOv8m | ULTRALYTICS

Final Comparative Report - Graduation Project

Andrew Wageh

April 19, 2026

Dataset: 12,218 X-Ray Images | Training: 5 hours

1

EXECUTIVE SUMMARY

This report compares two object detection models for automated hip and pelvis fracture detection in X-ray images: **Faster R-CNN** and **YOLOv8m**, both trained on a merged dataset of **12,218 X-ray images**. The YOLOv8m model significantly outperforms Faster R-CNN, achieving **79.7% mAP50** compared to **48.6% mAP50**, representing a **31.1% absolute improvement**. YOLOv8 also provides faster inference (~0.03s/image vs ~0.5s/image) and deterministic predictions, making it the recommended choice for the AETHEA medical platform.



2

DATASET OVERVIEW

The dataset combines hip and pelvis X-ray images from two sources, merged into a unified dataset with binary classification: **fracture** and **no_fracture**. Images were split into training, validation, and test sets while maintaining class distribution.

Splitting	Hips & Pelvis	Hips	Pelvis
Training	11,840	6,300	5,148
Validation	184	200	~180
Test	194	220	~190
Total	12,218	6,700	5,518

RESULTS COMPARISON

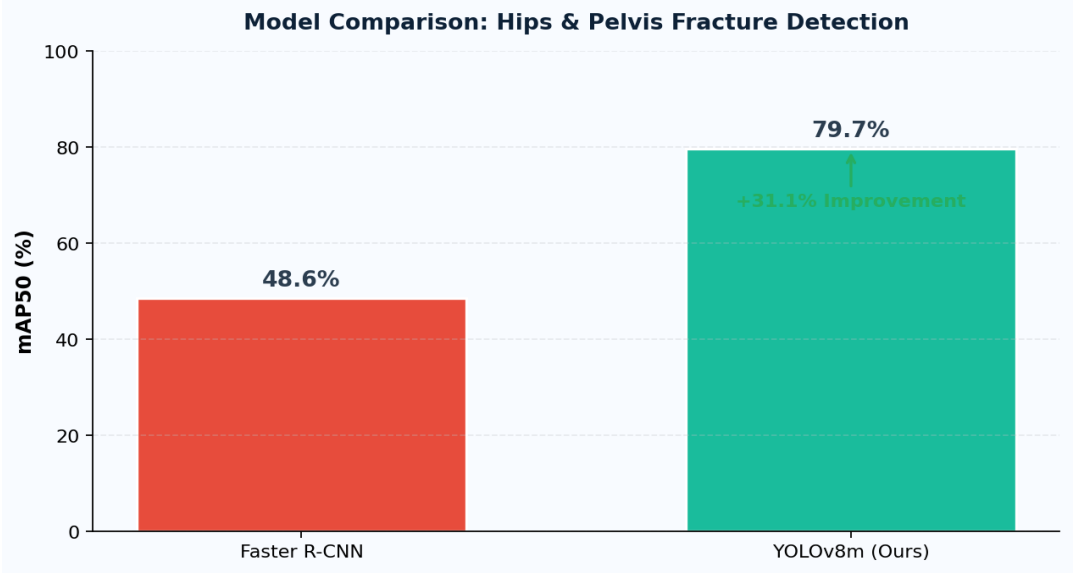


Figure 1 - mAP50 comparison: Faster R-CNN vs YOLOv8m.

Aspect	Faster R-CNN	YOLOv8
mAP50 (IoU=0.50)	48.6%	79.7%
AP (IoU=0.50:0.95)	45.60%	52.4%
Precision	62.3%	86.9%
Recall	51.9%	75.0%
Inference Speed	~0.5s/image	~0.03s/image
Prediction Consistency	■ Inconsistent	■ Deterministic
Training Epochs	5	38
Training Time	~3:30 hours	~5 hours

4

YOLOV8 DETAILED METRICS

The YOLOv8m model was trained for **38 epochs** on Kaggle with T4 x2 GPUs, completing in approximately **~5 hours**. Early stopping was triggered at epoch 38 as no improvement was observed for 20 consecutive epochs. The best model was saved at epoch 18.

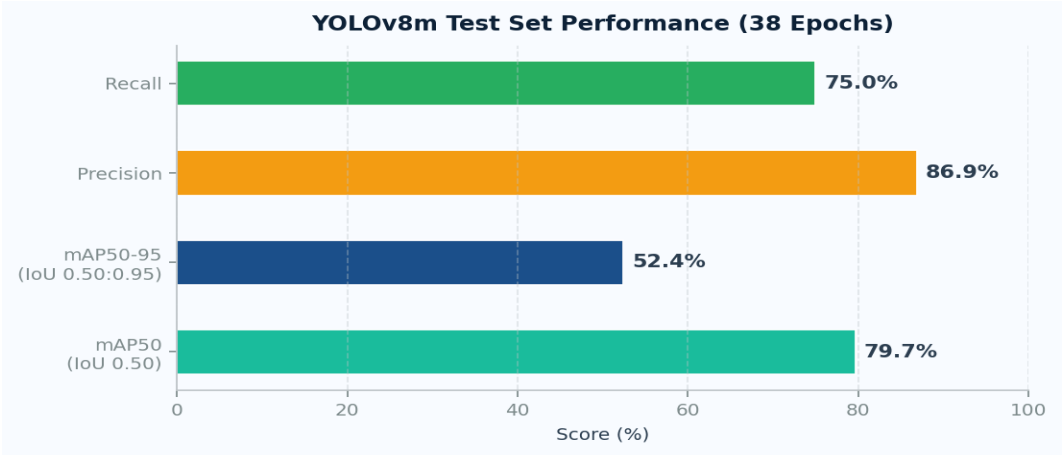


Figure 2 - YOLOv8m test set performance metrics.

5

CONCLUSION AND RECOMMENDATION

YOLOv8m is strongly recommended for production deployment in the AETHEA medical platform. It achieves **79.7% mAP50** (vs 48.6% for Faster R-CNN), a **31.1% absolute improvement**, while providing faster inference and perfect prediction consistency. The model's high precision (86.9%) and recall (75.0%) make it suitable for clinical-assist applications.

Andrew Wageh | AETHEA Graduation Project